

VUMMETHALA VENKATA SRI DATTA CHARAN

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PROFESSIONAL SUMMARY

Responsible AI Engineer and Generative AI Specialist with 1+ years of enterprise production experience building robust LLM safety systems, alignment frameworks, and automated multi-agent agentic workflows. Proven track record in orchestrating large-scale synthetic data generation pipelines and Supervised Fine-Tuning (SFT) over distributed high-performance GPU environments. Adept at creating real-time interception proxies and Model Context Protocol (MCP) implementations mapped to rigorous organizational, machine learning, and software security guardrails.

PROFESSIONAL EXPERIENCE

Infosys

Specialist Programmer (L2) — Responsible AI Engineer (AI Safety, Alignment & Guardrails)

Jul 2024 – Present

AI Guardrails, PII Privacy & Application Security

- Architected and deployed a standardized Model Context Protocol (MCP) Server acting as a boundary proxy to actively identify, log, and mitigate PII leakage, toxicity, jailbreaks, and adversarial prompt injection.
- Programmed low-latency proxy intervention layers using mixed rule-based and instruction-based models to catch, log, and block/warn about prompt injection vectors, jailbreak payloads, and toxic patterns.
- Designed highly specialized runtime filter hooks and rule-based/instruction-based privacy guardrails within corporate developer AI workspaces (Claude Code, GitHub Copilot) to intercept and mask sensitive Personally Identifiable Information (PII) before external API exposure.
- Developed granular code-safety validation mechanisms within developer assistant proxies that automatically inspect LLM output code blocks to isolate, warn, and block potentially destructive or irreversible execution commands.
- Mitigated critical open-source supply chain vulnerabilities in developer assistants by enforcing mandatory authorization sequences for third-party tools and verifying library source provenance against the organization's secure Artifactory registry.
- Implemented command-level explainability checks that force agents to generate comprehensive justifications, risk profiles, impact analyses, and action reversibility indicators prior to executing script commands.

Model Alignment, Supervised Fine-Tuning (SFT) & Safety Benchmarking

- Curated specialized alignment datasets and conducted continuous adversarial red teaming simulations on Slurm GPU clusters to identify structural safety gaps and eliminate specific bias and toxicity vulnerabilities.
- Executed Supervised Fine-Tuning (SFT) on Small Language Models (SLMs) using Unsloth across distributed environments, achieving a 34% increase in baseline safety benchmarks while preserving functional helpfulness.

Data Engineering, Inference Optimization & Agentic Workflows

- Accelerated data processing pipelines by wrapping fine-tuned systems with optimized vLLM configurations to drive high-throughput synthetic data generation and automated multi-turn alignment filtering.
- Engineered a programmatic Threat Simulator Suite for continuous, automated adversarial testing against targeted agentic applications to isolate logic bypass vulnerabilities.

- Architected an enterprise automated risk classification and parameter extraction engine for corporate Statements of Work (SOW) leveraging stateful multi-agent LangGraph workflows.

EDUCATION

**Indian Institute of Information Technology, Design and Manufacturing
(IIITDM Jabalpur)**

Bachelor of Technology (B.Tech.) in Computer Science and Engineering

Cumulative Performance Index (CPI): 7.9 / 10.0

2020 – 2024

PROJECTS & HACKATHONS

Shop AI — Hybrid Agentic E-Commerce Platform github.com/VvSdC/ShopAI

React.js, Node.js, Express.js, MongoDB, LLM AI Agents, JWT, Stripe, Cloudinary

- Engineered a full-stack e-commerce system blending a traditional storefront with a parallel natural language conversational AI Agent workspace capable of executing cart mutation and dynamic order processing.
- Designed a robust execution sandbox enforcing precise tool-calling separations to secure state mutations against unauthorized excessive agency and cross-account data exposure.
- Built an event-driven LLM metadata extraction pipeline that automatically generates search-optimized database tags alongside a twin moderation layer to filter out public review toxicity while preserving constructive critique.
- Secured commercial pipelines using a webhook-decoupled Stripe ingestion flow alongside stateless client state preservation via encrypted JSON Web Tokens (JWT) and high-availability Cloudinary CDN media delivery.

AEGIS — AI Ethics & Governance Intelligence System aegis-backend-6y5ofugcka-el.a.run.app

Python, Gemini CLI, GCP Cloud Run, YARA, Regex, SAST, JSON-RPC, REST, Semgrep, Bandit

- Built a production-grade AI gateway proxy sitting between client endpoints and foundational LLMs to implement real-time safety guardrails, policy verification checks, and short-circuit breaker compute optimizations.
- Developed a high-efficiency dual-tier interception logic combining an ultra-fast (<30ms) regex/YARA pattern screening pass with an asynchronous Gemini semantic analysis layer for complex zero-day exploits.
- Configured defenses covering 91 prompt injection topologies alongside a response-side SAST checker using 160 custom evaluation paths addressing OWASP LLM Top 10 vulnerabilities (including insecure code and logic bypasses).
- Programmed a dynamic structural compliance parser handling 105 distinct data categories mapped to verification rules across international data and system deployment boundaries.

TECHNICAL SKILLS

- **Core Engineering & ML:** Machine Learning, Deep Learning, NLP (Natural Language Processing), Data Structures & Algorithms
- **Responsible AI & Safety:** AI Guardrails, Prompt Injection Defense, Jailbreak Mitigation, PII Masking & Redaction (105+ entities), Toxicity Filtering, Red Teaming, Adversarial Testing, OWASP Top 10 for LLMs, Supply Chain Security
- **Model Alignment & Fine-Tuning:** Supervised Fine-Tuning (SFT), Unsloth, Small Language Model (SLM) Optimization, Safety Benchmarking, Dataset Curation, Synthetic Data Generation, Model Robustness Evaluation
- **Generative AI & LLM Frameworks:** Large Language Models (LLMs), LangChain, LangGraph, Model Context Protocol (MCP), vLLM, Gemini CLI, Agentic AI, RAG
- **Compute & Infrastructure:** NVIDIA H100/B200 GPU Clusters, Slurm Workload Manager, Distributed Compute, Google Cloud Run (GCP), Virtual Machines (Linux), Docker
- **Security & Data Libraries:** PyTorch, Scikit-learn (Sklearn), Pandas, NumPy, Matplotlib, Static Application Security Testing (SAST), YARA Rules, Regex Pattern Matching, JWT, Bandit, Semgrep

- **Languages & Full-Stack:** Python, JavaScript, React.js, Node.js, Express.js, REST APIs, JSON-RPC, MongoDB, Cloudinary
- **Tools & Platforms:** Git, Linux, Hugging Face, GitHub Copilot, Claude Code, Azure AI Services

CERTIFICATIONS & CREDENTIALS

- **Claude Certified Architect (Early Adopter)** | Anthropic
- **Microsoft Certified: Azure AI Engineer Associate** | Microsoft
- **Applied Generative AI Professional** | Infosys Certified